

SHRIVIDYA HEGDE

Senior AI Data Engineer | Apache Airflow Contributor & Champion | 5x Astronomer Certified | LLM / RAG Pipelines | Cloud-Native Data Platforms

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PROFESSIONAL SUMMARY

Senior AI Data Engineer with 10+ years building cloud-native data platforms, ETL/ELT pipelines, and production LLM/RAG systems. Apache Airflow Champion and open-source contributor (7 merged PRs; #1 Airflow Champion globally, Q3 2026), with valuable professional certifications. Architects production RAG systems end to end - LangChain, ChromaDB, and FAISS for retrieval; Ragas for evaluation; NeMo Guardrails and Guardrails.io for safety - across AWS, GCP, and Azure. NJ Chapter Lead for Women in Data, mentoring engineers and speaking at industry conferences.

SKILLS

Data Engineering: Apache Airflow 3, Astronomer/Astro, ETL/ELT Pipelines, Data Orchestration, Distributed Systems, Scalable Architecture, dbt, Data Modeling, Data Warehousing, Data Lakehouse, Apache Iceberg, Apache Spark, PySpark, Batch Processing, PostgreSQL, ClickHouse, Snowflake, Databricks, Streamlit

AI / ML Engineering: LLM Integration, RAG Pipelines, LangChain, OpenAI API, Agentic AI (CrewAI), Vector Databases (ChromaDB, FAISS, Weaviate), RAG Evaluation (Ragas), LLM Guardrails (NeMo Guardrails, Guardrails.io), NLP (NLTK, TextBlob, VADER), Sentiment Analysis, Semantic Search, Transformers, Prompt Engineering, Generative AI, MLOps

Cloud & DevOps: AWS (Certified), GCP, Microsoft Azure, Kubernetes, Docker, Terraform, CI/CD, GitHub Actions, CircleCI, GitLab CI, Azure DevOps, Infrastructure-as-Code

Programming: Python, Java, SQL, PySpark, Shell, FastAPI, GraphQL

Data Quality & Observability: Data Quality Engineering, Data Governance, Data Lineage (OpenLineage), Observability, Reliability Engineering, Production Monitoring, Performance Optimization

Testing Frameworks: Selenium, Playwright, Pytest, Appium, BrowserStack, REST Assured, Postman

Leadership & Collaboration: Technical Strategy, Team Mentoring, Cross-Functional Collaboration, Open-Source Governance, Stakeholder Communication, Conference Speaking, Technical Writing

PROFESSIONAL EXPERIENCE

Apache Airflow Open-Source Contributor | The Apache Software Foundation

Remote • Mar 2026 - Present

- Contribute to Apache Airflow core with 7+ merged PRs - code improvements, bug fixes, and documentation - to one of the most widely adopted data orchestration frameworks.
- Recognized as Apache Airflow Champion (Jan 2026) by Astronomer for community contributions, PR reviews, issue triage, and ecosystem advocacy; Top 5 globally in Q2 2026 and Q3 2026, and #1 Airflow Champion globally for Q3 2026.
- Contribute to Airflow 3 governance and feature development - DAG authoring, dynamic task mapping, and scheduler improvements.
- Review pull requests for Apache Spark as part of open-source community involvement.

Senior Software Engineer, Astro Observe (Data Platform) | Astronomer

Bengaluru, India (Remote) -> Parsippany, NJ (Nov 2024) • Mar 2022 - Nov 2025

- Joined as SDET; expanded scope to architect cloud-native data pipeline frameworks for Astro (Astronomer's managed Airflow platform) using Python, Airflow DAGs, and cloud SDKs (boto3, azure-sdk, gcp-sdk, astro-sdk) - throughput up ~30%.
- Designed and optimized ETL/ELT orchestration workflows with Airflow and Kubernetes across GCP, Azure, and AWS - data delivery timelines down 20%+.
- Architected an LLM-powered RAG recommendation system (LangChain, ChromaDB) that summarized test logs for release gating, flagged bugs, and routed findings to developers via Slack - productivity and resolution speed up ~40%.
- Built cloud resource lifecycle automation (GCP CLI, AWS CLI) - infrastructure costs down ~60%.
- Embedded data quality and observability checks into CI/CD (CircleCI, GitHub Actions), securing pipeline reliability and data integrity at enterprise scale.

Senior Test Automation Engineer, Core Payments Data | Commonwealth Bank of Australia

Bangalore, India • May 2021 - Feb 2022

- Led automation strategy for Core Payments data systems - 95% automation coverage, testing costs down 15%.

- Designed reusable automation frameworks across microservices environments for faster, more reliable pipeline validation.
- Awarded Best Partner for outstanding automation framework delivery.

Consultant, Automation Specialist | KPMG India

Bengaluru, India • May 2020 - May 2021

- Built automation frameworks across 4+ project teams - manual QA cycles down 25%, adoption up 30%.
- Ran a SAP S/4 HANA proof of concept (Eggplant Functional, DAI), delivering data readiness and governance recommendations.
- Extended mobile data validation across Android and iOS using BrowserStack and Appium.

IT Analyst, Automation Engineer | Tata Consultancy Services (TCS)

Sydney, Australia & Bangalore, India • Jun 2016 - Apr 2020

- Built unified automation frameworks for CI/CD across Banking & Finance data platforms - manual testing effort down 50%, deployment frequency up 35%.
- Engineered Selenium WebDriver, Java, and Python automation across Agile and Waterfall delivery models.
- Won the Service Commitment Award and On the Spot Award for critical project delivery.

Software Engineer / ASE Trainee | Tata Consultancy Services (TCS)

Bengaluru, India • Jun 2014 - Jun 2016

- Built Java automation tools for SQL/PL-SQL data operations - manual database intervention down 60%; trained in Informatica, BI, and PL/SQL, forming a foundational data engineering skillset.
- Won On the Spot Award (x3) and the Special Initiative Award (1st Prize, National ILP Debate Competition).

FEATURED PROJECTS

Resume Analyser RAG | LangChain, FAISS, OpenAI API, Python

- Built a LangChain + FAISS RAG pipeline answering natural language questions about uploaded resumes, with multi-document retrieval and semantic search.

OpenAI Sample Chatbot | OpenAI API, Python, Prompt Engineering

- Built a chatbot demonstrating advanced prompt engineering and OpenAI API integration, including system message design and context management.

Article Insights NLP Pipeline | Apache Airflow 3, Python, NLTK, TextBlob, NLP

- Automated an Airflow 3 NLP pipeline running parallel DAG tasks - keyword extraction, summarization, sentiment analysis, named entity recognition, and readability scoring - with structured JSON output for content strategy and SEO.

Real-Time Stock Sentiment Analyzer | Python, Streamlit, TextBlob, yFinance, Docker, GitHub Actions

- Built and fully Dockerized a real-time stock and news sentiment dashboard in Streamlit, with CI/CD via GitHub Actions and live/sample data modes for end-to-end MLOps deployment.

CERTIFICATIONS

- Astronomer Champions Program - Apache Airflow | Astronomer | Jan 2026
- AI Engineer for Developers Associate | DataCamp | Jul 2026
- Data Engineer Associate | DataCamp | Apr 2026
- DAG Authoring & Fundamentals - Airflow 3 | Astronomer | Jun 2025
- AWS Certified Cloud Practitioner | Amazon Web Services | Jun 2025
- DAG Authoring & Fundamentals - Airflow 2 | Astronomer | Mar-May 2025
- Generative AI Fundamentals | Databricks Academy | Mar 2025
- Additional: Python Data Associate, AI Fundamentals, Python for Data Science (DataCamp) | Java Certified (HackerRank) | Certified Selenium Professional (Vskills)

COMMUNITY, SPEAKING & LEADERSHIP

- Women in Data (Trademark) - NJ Chapter Lead (Mar 2025-present): run bootcamps on SQL, Python, PySpark, and applied ML; mentor women across multiple organizations via informal 1:1s (Topmate) on Data Engineering, Apache Airflow, and orchestration challenges.
- Spoke at Apache Airflow NYC MeetUp (Jun 2026) on "Airflow + AI: Orchestrating the Intelligent Stack"; session accepted at Apache Airflow Summit 2026.

- Technical author on Medium and Substack - 6+ articles on RAG, LLM agent deployment, and Airflow cost optimization; Expert Content Writer for Data Engineering Byte.
- Podcast guest on The DataFlow Cast (Kenten Danas) and Ediso Conversations; speaker at Future World Conversations/Alliance; nominated for the Women Tech Network Global Awards.

EDUCATION

Bachelor of Engineering, Computer Science & Engineering

Malnad College of Engineering | CGPA 9.06